



Enhancing Academic Writing Integrity: Ethical Implementation of Generative Artificial Intelligence for Non-Traditional Online Students

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Abstract

Despite continuous discussions on Generative Artificial Intelligence (Gen AI) ethics in academia, limited studies examine the perceptions of students or the application of ChatGPT with non-traditional students. We report an exploratory single case of a two-week orientation offered to non-traditional students entering online bachelor's degree programs. Students that participated in the orientation course completed a survey and a "short research-based reflection paper" using the pre-developed prompt discussing the relationship between academic dishonesty and workplace ethics including the ethical use of Gen AI. Based on open coding of 63 selected essays, we identified four major themes in student reflections on the ethical use of Gen AI: (1) threats to integrity and the learning process; (2) benefits and efficiency; (3) perceived inevitability of Gen AI; and (4) shared responsibility to govern the ethical use of Gen AI in academic life. The purpose of this study was to provide insights to researchers and practitioners in higher education on designing writing activities that can alleviate threats to academic integrity while promoting benefits of Gen AI into the classroom and workplace.

Keywords Academic writing · Artificial intelligence · Ethics · Generative artificial intelligence · Integrity · Non-traditional students · Online learning

Non-traditional students have become prevalent in many institutions in the United States, with at least 70% of undergraduate students in the United States meeting at least one criteria to be considered non-traditional (NCES, 2015). A widely used definition of a non-traditional student is someone that is over the age of 24, who often has family and work responsibilities (NCES, 2022). Many non-traditional students enrolled in online degree programs benefit from the flexibility of online courses maintaining their employment or family responsibilities (Appana, 2008). While non-traditional students often bring valuable workplace skills into the classroom, they also often experience gaps in knowledge regarding the use of technology, academic writing, professional integrity, and research ethics (Safford & Stinton, 2016).

Examples of Generative Artificial Intelligence (Gen AI) available in the market include Co-Pilot, Claude, Gemini, and ChatGPT. Among many Gen AI tools, ChatGPT most widely used and known to public has become a buzz word since its public launch in 2022 by OpenAI (OpenAI, 2022). It is considered an evolutionary next step to chatbots, which give responses when a prompt is entered using natural language processing model. Although it is exciting to experience a rapid advancement of AI-assisted tools, research communities have voiced concerns related to ethical uses of Gen AI in academic writing.

Many higher education institutions and publishers have created taskforces with experts in AI, academic writing, and academic integrity to establish appropriate ethical standards for the use of AI within academia. For example, publishers including *Springer*, *Wiley*, *SAGE*, *Emerald*, and *Frontiers* have developed their own standards and announced their stance on allowable use of Gen AI's in writing academic papers for their journals. For example, *SAGE* highlights that authors have full responsibility in checking the factual accuracy, validity, and appropriateness of content created by Gen AI technology (Sage, 2023). *Wiley* emphasizes that transparency is key and

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that authors need to identify where Gen AI tools have been used within the submitted paper (Streeter, 2023). Also, authors are encouraged to report it in the acknowledgments section of the manuscript if applicable. Additionally, many journals are members of organizations such as the Committee on Publication Ethics (COPE), World Association of Medical Editors (WAME), and the JAMA Network, which have banned the use of ChatGPT as a coauthor in research papers.

Purpose Statement

Although there has been much discussion in higher education on the ethical integration of Gen AI in course assignments and research papers (e.g., Haleem et al., 2022; Holden et al., 2021), limited studies examine its application with non-traditional students in an online learning environment. This research seeks to inform our understanding of the non-traditional student perceptions of Gen AI and the tools and resources that non-traditional students need to successfully navigate this new technology in their academic journey as well as inform the design of instruction that supports the level of student's technology proficiency. In this way, instructors and administrators in higher education can find ways to better support these non-traditional students to succeed in the era of artificial intelligence. This research also seeks to add to the small number of empirical studies which have examined student perceptions of ethical considerations in using AI in their course assignments, in their academic field, and in their future workplace through the review of student's written work.

In this original single case study, we administered surveys and analyzed 63 essays submitted as part of a two-week orientation offered to incoming non-traditional students of online bachelor's degree programs in nursing, business, and liberal arts at a large university in the south-central region of the Southern United States. We aim to provide insights to fellow researchers and practitioners interested in academic integrity and the ethical considerations of using Gen AI in writing, particularly among the growing population of non-traditional students. To achieve this, we distributed surveys to understand their perceptions and understanding of Gen AI and analyzed student essays to identify student perception themes and develop suggestions on ethical issues implementing Gen AI for writing activities in a higher education context.

Non-Traditional Students and Technology

Prior research illustrates that non-traditional students may struggle more with technology adoption (Safford & Stinton, 2016), technology confidence (Eichelberger & Imler, 2016), and may perceive academic integrity policies and practices differently than their traditional counterparts (Harris et al., 2020). Digital natives were born and raised with technology,

making them typically familiar with computers and open to new advancements. While the term 'digital natives' typically describes most traditionally aged students, a significant portion of non-traditional students do not fit this description. As these individuals are not digital natives, their exposure to new technology occurs in a different manner compared to their traditional counterparts (Eichelberger & Imler, 2016). As non-traditional students experience more challenges than their traditional student counterparts with learning new technology (Safford & Stinton, 2016), understanding their perceptions toward Gen AI may differ as well.

Generative Artificial Intelligence in Educational Research

Although ChatGPT is a relatively new tool, in the past year, many studies have already been published through special issues and AI related books. Some studies have focused on learning the potential benefits and drawbacks of Gen AI in teaching and learning (Baïdoo-Anu & Ansah, 2023; Haleem et al., 2022; Tlili et al., 2023). Most were exploratory studies sharing ChatGPT's responses on their research questions (Baïdoo-Anu & Ansah, 2023; Cooper, 2023), and literature reviews (Haleem et al., 2022; Kabudi et al., 2021; Korzynski et al., 2023). In a qualitative case study, Tlili et al. (2023) examined early adopters of ChatGPT to learn about their concerns of its use in educational settings. Their study revealed that developing reasonable educational chatbots that promote fairness and appropriate use of personal data is pertinent.

While using ChatGPT may have great advantages in deepening student's learning on certain topics, there is also concern that it may harm the learning process. Regarding the trustworthiness of the information used by ChatGPT, it may exacerbate students' negative behaviors such as dishonesty, manipulation, and misinformation (Tlili et al., 2023). Most chatbots are task-oriented (Skjuve et al., 2022), using machine learning and deep learning techniques to generate text (Korzynski et al., 2023). Therefore, when ChatGPT is unethically used, it may lead to human unintelligence and unlearning (AlAfnan et al., 2023). Experts also perceive a likelihood of reducing students' critical thinking (Tlili et al., 2023) and intentional and unintentional plagiarism including direct copy-pasting of generated text to student assignments.

Strategies for Using ChatGPT for Writing Assignments

With an appropriate prompt, ChatGPT can generate virtually any text using language processing technology. What differentiates ChatGPT from a search engine is the ability to continue a conversation with follow-up questions and comments (Fitria, 2023). One of the key features of ChatGPT is its ability

to process vast amounts of textual information in a short period of time (Dergaa et al., 2023). As this advantage may harm student's learning, researchers have begun to develop strategies to avoid students unethically utilizing ChatGPT for academic writing, writing assignments, and take-home exams. For instance, AlAfnan et al. (2023) recommend four strategies for instructors using ChatGPT for writing, which include: (a) refrain from making theory-based questions for take-home assignments; (b) provide detailed case-based and scenario-based assignment tasks that call for personalized answers; (c) enforce the use of plagiarism detection software; and (d) integrate ChatGPT generated responses as examples in classes. In addition, Fitria (2023) recommends that students handwrite or use typewriters to avoid cheating on written assignments and exams. In addition to unethical writing practices, other researchers have raised concerns that ChatGPT hinders student's critical thinking skills (Tlili et al., 2023) such as their ability to think and write without technological assistance (Fitria, 2023).

Other researchers (Su et al., 2023) argue that features of ChatGPT may assist in students' argumentative writing development by supporting tasks like outline preparation, content revision, editing feedback, proof-reading, and post-writing reflection. Second language learners may also benefit from ChatGPT, receiving help with language-related feedback, proofreading their written work, and translating difficult text (Su et al., 2023).

Non-Traditional Students and Technology Proficiency

Previous research indicates that non-traditional students experience difficulties juggling academic work, full and part-time jobs, and caregiving duties (Safford & Stinton, 2016). Taking online courses may pose an additional burden for non-traditional students compared to those attending in-person classes, as they must learn to use various technologies to participate remotely (Deng & Zheng, 2021; Safford & Stinton, 2016). In this study, non-traditional students enrolled in an online bachelor's degree program at a large institution in the Southern United States had reported relatively low awareness of ChatGPT or Gen AI with 58% ($n=38$) of students indicating they have heard of ChatGPT or Gen AI and only 26% of those students ($n=10$) had experience using it. Through an analysis of essay assignments completed by non-traditional students, this paper suggests ways for educators in the field to integrate Gen AI in various educational contexts appropriately. The research questions that guided our inquiry were:

- To what extent do non-traditional online students have experience with using Gen AI tools?
- What factors do non-traditional online students perceive as critical for the ethical use of Gen AI?

- How do non-traditional online students view their responsibility in maintaining academic integrity when using Gen AI?

Methods

We employed an exploratory single case study method (Yin, 2014) to examine student's understanding towards ethically integrating Gen AI in their courses in an online bachelor's degree orientation at a large Southern University in the United States. An exploratory case study aims to investigate a phenomenon in a specific context, using one or more data collection methods. Typically, the study answers "how," "why," and "what" questions (Yin, 2014). In this study, the cohort of incoming non-traditional students of an online bachelor's degree program was selected as a single unit of the case. Prior to conducting the study, we received approval from the institutional ethics review (IRB Protocol # 2,306,477,368).

Participants and Study Context

As a part of an orientation course in early August of 2023, incoming non-traditional students for an online undergraduate program in business, nursing, liberal arts, and humanities completed a "short research-based reflection paper" discussing the relationship between academic dishonesty and ethical use of Gen AI related to their field of study. We analyzed the students' papers using an open coding method to identify themes (Creswell, 2017) in their articulated perceptions of the ethical use of Gen AI in educational settings and beyond.

Participation in the orientation course is considered mandatory for students entering online undergraduate programs at the institution. However, students are not penalized for non-participation. The course takes place the two weeks directly preceding the students first major term of study in that program and is designed to orient students to the technology, norms, and the common structure of online courses offered by the institution in a highly condensed format. Students are assigned to orientation sections based on their majors. There they interact with their classmates and a trained facilitator while completing the assigned coursework. The research reflection paper meets multiple course learning objectives including navigating the use of university resources like the library and the academic writing center.

Most students in online undergraduate programs at the institution meet multiple common markers of non-traditional students (Eichelberger & Imler, 2016). Less than one percent (0.07%) of students are traditionally aged (17–20) with an average reported age of 31. The majority of students (63%) are enrolled part time and more than 90% indicate being

employed either full-time (78%) or part-time (16%) while attending school. Ninety-one percent of the students reside in the southern United States and 55% reside in the same state as the institution. Students classified as online are enrolled in one of thirteen majors available online which range across disciplines including those in business, nursing, liberal arts, and humanities.

Data Collection and Procedure

The study underwent a rigorous and meticulous planning process from a pilot study as a needs assessment on understanding the previously offered essay assignment on academic integrity (without AI component). To aid in the development of the prompt for the study, a review of the writing assignment previously used in the orientation course was conducted to support the development of an initial coding framework and assist in the design of the prompt used in this study.

In Spring 2023, a preliminary analysis of papers used the existing assignment with a prompt addressing the relationship between academic integrity in higher education and ethical behavior in the workplace (see Fig. 1). However, this paper did not ask students to consider addressing AI as Gen AI was not as prevalent then. A preliminary academic integrity theme codebook was developed from the sample of prior work. Results from that study were also used to develop the new prompt, which extended the student assignment to also consider the possible role of Gen AI in academic writing and integrity in both their academic and career journeys. Results from that study were also used to develop the new prompt which extended the student assignment to also consider the possible role of Gen AI in academic writing and integrity in both their academic and career development journeys.

In August 2023 students who participated in the two-week orientation course for new online undergraduate were assigned to complete a “short research-based reflection paper” using the developed prompt discussing the relationship between academic dishonesty and workplace ethics including the ethical use of Gen AI related to their field of study. Of the 72 papers submitted, 63 of the assignments submitted met the qualification for inclusion in this study. Requirements for inclusion in the study were that the paper

was submitted by a student in the August 2023 orientation course, the paper could be effectively de-identified, and student addressed the Gen AI portion of the assignment. Those papers that did not address this portion of the assignment were excluded from this analysis, including eight that addressed academic integrity but not Gen AI and one that addressed neither of the prompt points.

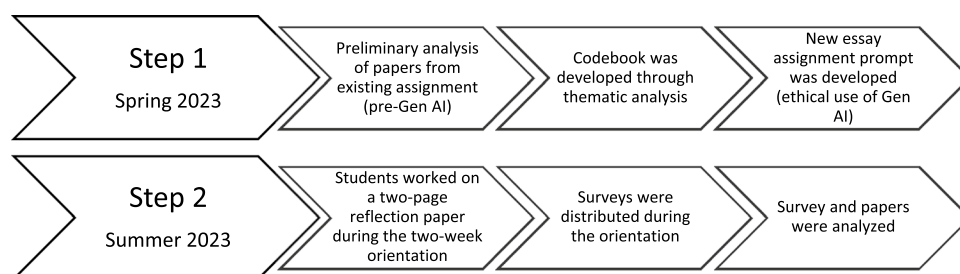
Papers were downloaded from the learning management system by program staff not associated with the study and de-identified prior to being provided to the researchers. Student names and sections were removed from the papers, but markers within the papers that could be used to identify the major of the student were not removed as they were relevant to the content of the paper themselves and each major group (e.g., business, nursing, liberal arts, humanities) contained enough students to adequately maintain the anonymity of the students' submitting papers. Deidentified papers were then transmitted to the researchers electronically via a secured drive.

Data Analysis

Based on the article choices and the content of the essays the reviewers were able to determine the likely major of 95% ($n=60$) of the students. Business was the most common major represented (46%) followed by Humanities (25%) and Nursing (24%). Papers were reviewed against the codebook developed using the prior papers (Creswell, 2017) and the codebook was extended to reflect the new Gen AI content (see Table 3). Themes specific to Gen AI were constructed using an iterative open coding process by which common ideas were identified and collected through the review of the writing samples. Papers were reviewed using a manual paper and pencil method in random order in an iterative process coding both convergent and divergent concepts. Sections of the paper corresponding with the identified codes were highlighted and direct quotes retained to an additional document identified by the code and the identifier assigned to the paper for tracking purposes. Upon completion of initial review of the papers the codes were reviewed and condensed to establish the themes presented in this paper.

After the initial set of themes were constructed, the full set of papers were reviewed again to determine if themes accurately reflected both the convergent and divergent

Fig. 1 Data collection procedure



perceptions articulated in the papers and the final themes constructed and addressed in the results. The descriptive analysis was conducted using Microsoft Excel. The percentage of the papers in which each theme was coded is presented in the findings indicating the prevalence of each theme. To ensure the trustworthiness of the data, the authors conducted multiple rounds of reviews until reaching a consensus on the emerging themes, facilitating investigator triangulation. We also analyzed data from multiple sources (e.g., survey, document analysis of 63 essays) for data triangulation. The survey administered at the beginning of the course prior to the paper assignment included questions related to perceptions and experiences using Gen AI tools, and demographic information. We analyzed 63 essays that are 1.5-to-5-page length.

Generative Artificial Intelligence Writing Assignment Instructions

Students in the August 2023 cohort were given the prompt developed through the preliminary study that included examination of the role Gen AI might play in academic integrity in both academic and workplace settings. In their short paper, they were directed to review three peer reviewed articles available through the university library system and discuss what each article said about the relationship between academic integrity and workplace ethics in their field of study and the role of Gen AI in academic integrity and writing.

Students were provided with a list of six studies that were hand-picked by the two researchers of this study. The papers were selected using the search term of “Generative AI in education,” and “AI in education.” Admittedly, there could be limitations of providing the students with articles as it may potentially influence students’ perceptions towards Gen AI. However, when we conducted the orientation in the summer of 2022, we believed that providing students with a list of articles was the best option, given the limited number of publications on Generative AI available at that time.

As described in Table 1, students were instructed to independently identify two scholarly articles using the university library resources related to academic integrity and ethics in their chosen field and then directed to include at least one article on Gen AI. Students were provided with a list of articles they could choose from related to Gen AI or had the option to identify another article that met the assignment parameters of a peer reviewed scholarly source available through the university library which addressed the emerging role of artificial intelligence in education. Eighty percent ($n=50$) of the students chose to use articles on the list, six percent used a provided article and independently identified article, and 8% used only independently sourced

articles. Six percent cited no specific articles on Gen AI but still expressed opinions about its use.

The purpose of providing a list of resources was to support the learning process of identifying reliable peer-review articles from the library which was an established primary learning outcome for the assignment. Based on the timing of the study (Summer 2023), there were a limited number of high-quality peer-reviewed educational journal articles available through the institution's library as the public launch of ChatGPT was less than a year ago. Providing a list for students to use allowed us to ensure the accessibility of articles that would meet the standards required for the assignment. While efforts were made to ensure a variety of perspectives were addressed in the articles, the authors acknowledge possible bias in the selection of the articles as a limitation of the study, not many high quality peer-reviewed education educational articles on Gen AI and academic integrity were published at the time.

Study Findings

General Academic Integrity

Using the codebook established in the Spring 2023 review of the writing samples, the presence of the five established outcomes were identified in the writing samples addressing the relationship between academic integrity and workplace ethics as well as a perception of responsibility or control over maintaining academic integrity. The established codebook included five common outcomes: (1) the identification of at least one type of academic integrity violation, (2) articulation of risk to the academic process or learning by engaging in academic dishonesty, (3) articulation of negative impact on the workplace of academic integrity or the articulation of a relationship between academic dishonesty and workplace dishonesty, (4) articulation of a personal connection to or role in preventing academic dishonesty, and (5) articulation of confidence to avoid academic dishonesty or uphold academic integrity.

As described in Table 2, 85 percent of students identified at least one type of academic integrity violation, with little variance between the majors. This finding was very consistent with the preliminary study in which 80% of the students were able to successfully identify at least one academic integrity violation type. Traditional examples of plagiarism were the most commonly referenced academic integrity issue identified, but unauthorized collaboration, use of unauthorized materials, and fabrication of data or research were also mentioned by students. Examples referencing Gen AI generally focused on wholesale paper fabrication but did include examples of unauthorized assistance in whole or in part of a variety of writing-based assignments.

Table 1 Essay prompt on academic dishonesty and workplace ethics**Assignment Description**

Write a minimum 1.5-page paper that discusses the relationship between **academic dishonesty and workplace ETHICS related to your field of study, and the role of Generative artificial intelligence (AI) in academic writing and integrity**. Your essay will consist of three peer-reviewed journal articles. First, choose two reliable, peer-reviewed articles from the library, about academic dishonesty and workplace ethics related to your selected field of study. For your third reading, choose one article from the list below (or do an independent library search) which addresses the emerging role of artificial intelligence in academic integrity.

Draw conclusions about the most common incidents that occur in your field. What conclusions does each of your sources come to? Write an introduction that includes a **summary** of the common themes in the articles. Then, for each source, write a paragraph that summarizes the findings and explains the significance to your field. Based on the third reading on Generative AI, you may write about the following topics: (1) potential benefits and challenges of AI-generated content in academia and the workplace; (2) strategies to tackle ethical issues in using AI-generated content in academic writing such as plagiarism, academic misconduct, copyright concerns, and authenticity of author's voice; and (3) the future of Generative AI in academia and the workplace. In the conclusion, articulate what you learned from the reading that is important for professionals in your field and what the additional challenges that students like you might face in maintaining ethical standards with the availability of artificial intelligence.

Purpose

This assignment introduces you to searching the library databases in order to locate a variety of articles about your topic as described above. You will then choose two of the best articles from the library and one article from the list provided related to artificial intelligence and summarize each in order to draw conclusions about the topic.

1. Locate several articles about academic dishonesty AND workplace ethics related to your selected field of study from library databases by searching the University of Arkansas Libraries. Ideally you are looking for articles that study or discuss possible relationships between academic dishonesty and workplace ethics though some articles may lead more towards one topic or the other, you want your collection of chosen articles to cover both concepts effectively
2. You may do your own search in the library database on Generative AI (i.e., ChatGPT) or choose one article from the list below regarding artificial intelligence and academic integrity
3. After reading the articles construct a paper summarizing each article and reflecting upon how the article relates to the topic using APA citations and style

Formatting

Use APA (7th edition) format:

1. Double spaced
2. 12 size font in Times New Roman
3. Minimum 1.5 pages to maximum 5 pages not including references page

Evaluation Rubric:**Intro and Conclusion (4 points)**

Your paper contains both introduction and conclusion paragraphs that are topical and highlight common threads and implications for your future practice.

Three Academic Articles (8 points)

Your paper references three academic articles or sources; Two from the library and one from the list provided

Article Summaries (12 points)

Your paper contains three summaries of three different sources that introduce the title, clearly articulate the main idea of the article without personal bias and show the significance of the article to your chosen field. Two are chosen from the library and one from the list provided.

APA Citations (8 points)

Your paper gives credit to your sources and includes proper APA style in-text citations throughout your paper and includes an APA style references page.

APA Format (4 points)

Your paper is correctly formatted in APA style formation (i.e. Title page included, header and page numbering at top right, running head, title of paper, student's name, college name double-spaced, 12 point font etc.).

Spelling and Grammar (4 points)

Your paper contains proper spelling, punctuation, and grammar.

Suggested Paper Structure:**Introduction****Body**

Summary 1—Article title

Summary of Article 1

Summary 2—Article title

Summary of Article 2

Summary 3—Article title

Summary of Article 3

Table 1 (continued)**Conclusion****References page****Helpful hints:**

- On the reference page articles should be listed in alphabetical order by first author's last name
- Follow APA 7th edition format for in-text citations and reference list
- Suggested search topics: academic dishonesty, academic integrity, ethics in the workplace, cheating, consequences, safety (any of these terms) + your chosen field of study or career options upon graduation
- Make sure your introduction discusses your initial thoughts about the topic and that your conclusion includes what you have learned through your research and from your selected articles

List of Chat GPT Articles:

- Chen, L., Chen, P., & Lin, Z. (2020). Artificial intelligence in education: A review. *Ieee Access*, 8, 75,264–75,278. <https://doi.org/10.1109/ACCESS.2020.2988510>
- Hwang, G-J., Xie, H., Wah, B. W., Gasevic, D. (2020). Vision, challenges, roles and research issues of artificial intelligence in education. *Computers and Education: Artificial Intelligence*, 1(100,001), 1–5. <https://doi.org/10.1016/j.caeai.2020.100001>
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- Tlili, A., Shehata, B., Adarkwah, M. A., Bozkurt, A., Hickey, D. T., Huang, R., & Agyemang, B. (2023). What if the devil is my guardian angel: ChatGPT as a case study of using chatbots in education. *Smart Learning Environments*, 10(1), 1–24. <https://doi.org/10.1186/s40561-023-00237-x>
- Zhang, K., & Aslan, A. B. (2021). AI technologies for education: Recent research & future directions. *Computers and Education: Artificial Intelligence*, 2(100,025), 1–11. <https://doi.org/10.1016/j.caeai.2021.100025>

Table 2 General academic integrity themes

Construct	Likely Major				
	Mean* (n = 63)	Nursing (n = 15)	Business (n = 29)	Humanities (n = 16)	Preliminary Study (n = 15)
Identify academic integrity	85%	80% (12)	86% (25)	87% (14)	80%
Importance to academics/learning	73%	80% (12)	75% (22)	62% (10)	53%
Importance to workplace integrity	78%	86% (13)	82% (24)	62% (10)	100%
Personal connection/ responsibility	52%	67% (10)	48% (14)	43% (7)	73%
Expressed confidence to avoid	13%	6% (1)	17% (5)	12% (2)	33%

*Unable to determine likely major for 3 papers. Disaggregated results not presented for this group due to small sample size

A larger portion (73%) of the August 2023 students articulated academic integrity as a threat to their own learning or academic process than in the initial study (53%) with nursing students (80%) the most likely to express the most concern followed by business students (75%). The most common academic threat identified for engaging in academic dishonesty was failing to learn (15) or understand the materials. Four students mentioned concerns about the impact of widespread academic dishonesty on the perceived value of education while others noted concerns about demotivating learning, limiting creativity, and perpetuating systemic oppression. Two students specifically noted that they were not convinced there was a negative impact on the academic process and that the current standards were outdated.

Seventy-eight percent of students articulated academic dishonesty as a threat to future workplaces, fewer than did

so in the preliminary study (100%). Nursing students (86%) were the most likely to articulate a workplace threat followed by Business students (82%). Specific threats expressed included compromising safety or operational standards that would result in negative outcomes for the employee, employer, or client/patient. Lowered motivation to succeed, limitations on creativity and innovation, and a lack of confidence in one's own abilities were also mentioned as threats. As with academic threats, humanities students (62%) identified these threats less often than business and nursing students.

Fifty-two percent of students articulated a personal role or responsibility in avoiding academic dishonesty, which was 21% less than in the preliminary study. On this indicator, nursing students (67%) expressed this commitment at a higher rate than did business (48%) and humanities (43%)

students. Very few (13%) of students in any major articulated a confidence to avoid academic integrity violations in their essays.

Generative Artificial Intelligence Themes

As reflected in the reviewed papers, student perceptions of Gen AI in academic writing and integrity could be categorized into four significant themes: (a) threats to integrity and the learning process, (b) benefits of Gen AI, (c) a perceived inevitability of Gen AI, and (d) shared responsibility to govern the ethical use of Gen AI in academic life. The most commonly articulated themes were perceived threats and benefits of the use for Gen AI in educational and workplace settings. In this analysis, threat types were grouped together as were benefit types as shown in Table 3 representing the percentage of papers in which each theme was identified. Beyond direct threats and benefits, two other themes were constructed: (1) perceived inevitability of Gen AI in education, and (2) shared responsibility for accountable and appropriate use of Gen AI including the need for academic integrity policies to adapt to new technology so that students and faculty clearly understand expectations and boundaries.

Mixed Perceptions

More students identified specific threats to academic integrity (79%) than articulated benefits (62%), most students articulated both benefits and threats (62%). Business students most frequently articulated threats, while nursing students most often articulated benefits (73%). Additionally, business students much less frequently articulated both threats and benefits (55%) compared to nursing (73%) and humanities (69%) students. Notably, 21% of students articulated elements of the technology or its use while assigning neither obvious threat nor benefit to the use of the technology as interpreted on the rubric. Though students appeared more likely to ascribe

threats than benefits after completing the assignment, the majority (62%) addressed both threats and benefits in their papers.

Threats to Integrity and the Learning Process

An increased prevalence of cheating and the ease of that cheating was the most commonly expressed threat (63%) of Gen AI. Students frequently articulated concern regarding total fabrication of an assignment, but also often indicated they viewed the use of Gen AI to replace portions of an assignment or a process as cheating. The prevalence of cheating as a threat was relatively similar across the majors, with Business students (72%) expressing it most frequently, followed by Nursing (67%) and Humanities (63%). A student expressed concerns not only for the increase in frequency but for the temptation to use Gen AI because of its ease of use.

With ChatGPT, you can give it a prompt, and it would generate what was requested within seconds, faster than completing the documentation or assignment yourself. As Artificial Intelligence continues to grow rapidly in popularity, I think the temptation to utilize this to get ahead in life and school becomes all too prevalent. Like with any new tool or application that could be deemed beneficial, will be used and abused for the wrong reasons.

Nursing students were more likely to express the risk as leading to negative workplace outcomes, specifically negative care outcomes.

AI as a tool for students runs the risk of nursing students using this tool to generate work for them without a full understanding of the material presented to them. This could potentially be hazardous as it becomes used more and more in the real-world environment as the nurse would not have the knowledge to double check the information generated for error before practicing on it.

Table 3 Generative AI themes

Major				
Construct	Mean	Nursing	Business	Humanities
Noted threats of Gen AI	79% (50)	73% (11)	86% (25)	69% (11)
Increase/ease of cheating	68% (43)	67% (10)	72% (21)	63% (10)
Inaccuracy of results	33% (21)	47% (7)	28% (8)	31% (5)
Threats to sense of humanity/integration	29% (18)	33% (5)	28% (8)	25% (4)
Privacy and consent concerns	8% (5)	13% (2)	7% (2)	0% (0)
Noted benefits	62% (39)	73% (11)	55% (16)	69% (11)
Addressed both threats and benefits	62% (39)	73% (11)	55% (16)	63% (10)
Inevitability of Gen AI	19% (12)	7% (1)	31% (9)	13% (2)
Shared responsibility to address implications of Gen AI	25% (16)	13% (2)	34% (10)	25% (40)

A substantial number of students also articulated concerns about the inaccuracy of Gen AI (33%) and the possible impact of human interaction (29%). In both cases, Nursing students articulated these concerns more often than business and humanities students though the variance was substantially more for concerns about inaccuracy (16–19% more common) and for concerns about the impact on human interaction (5–8% difference).

In addition to the obvious concerns of plagiarism and academic fraud, Gen AI has been known to be prone to hallucination. The AI will generate text that is not scientifically sound but sounds correct to the untrained or inattentive eye. This lack of true intellect is one of the main downfalls of Gen AI in the healthcare field.

Smaller, but still notable numbers of students identified Gen AI as a possible threat to privacy (8%) and raised concerns about bias (5%). Two students in Nursing and two students in Business as well as one student whose likely major could not be identified each addressed privacy related concerns in their papers. Specifically, students were concerned about consent for personal information to be included as part of the datasets being used by Gen AI.

“As of this moment there is no real way to consent to having your research used to train Gen AI and therefore its use automatically constitutes both plagiarism and theft.”

“Perhaps the most glaring is the issue of data privacy. Great care must be taken when using AI in regards to sensitive information such as legal information or medical records that might be protected by HIPAA.”

Additionally, three students, all of whom were identified as likely business students expressed concerns about bias in Gen AI rooted in the bias of Gen AI creators in the type of content used in their models.

Benefits of Gen AI

Sixty two percent of students identified possible benefits in the use of Gen AI both in the classroom and in workplaces. Nursing students (73%) were the most likely to address benefits followed by Humanities students (69%). Business students (55%) articulated possible benefits least frequently. Benefits commonly articulated by nursing students included workplace efficiencies particularly related to maintaining patient records as well as possible applications for assisting in diagnosis and the building of treatment plans.

“There are also potential benefits to Gen AI by using its capabilities to generate reports from select groups of data, gear questions to specific students, and predict treatment plans.”

As noted above, Nursing students (73%) more often noted both benefits and threats together. Nursing students who noted benefits also noted threats, particularly the risk of inaccuracy (47%) in Gen AI results and threats to human interaction and lack of perceived human care or concern.

Students also frequently noted possible benefits to the educational process itself through the use of Gen AI. Of particular interest to students was the potential of personalized learning and support that could be targeted to each learner more efficiently.

“Some methods that would be beneficial for nursing students is using ChatGPT to generate case studies to help prep for their exams.”

“This adaptability would allow Gen AI to gear questions and assignments towards the individual students needs based upon the questions that is asked of it.”

Additionally, some students also pointed out that in some cases the use of Gen AI might decrease some incidents of cheating by increasing efficiency of access to key knowledge and support.

The main goal of using chatbots or artificial intelligence seems to be that it is easier and quicker to get the responses needed. Such as, if a student has a question, they need an answer to right away, they could ask the chat bot instead of waiting for a human response. This could help the student to complete his or her work faster and they may not feel as stressed and resort to cheating. AI Chatbots can be helpful when they work properly.

While students identified the use of Gen AI as positive for instructional quality, this was an area in which some students expressed concern about potential bias and inaccuracy particularly when Gen AI would be used to evaluate student work.

The question becomes is it hypocritical for an institution to use it but not allow students. Yes, AI can make it easier to challenge academic integrity. But you cannot let one side of the population use it and not the other.... If there are harsh academic integrity policies that allow for no gray area, then teachers should not be allowed to use AI for their gray areas.

Students specifically articulated a perception of unfairness for instructors to be able to use Gen AI for efficiency if students were not given that same leeway.

Inevitability of Gen AI

Many impressions of the future implications of Gen AI in academic and workplace settings were articulated. Nineteen percent of students specifically noted the inevitability of the impact of Gen AI on their academic and career lives.

Chatbots, like ChatGPT, are not going away. The rampant growth of students utilizing these tools for various purposes in education cannot be, and has not been, ignored.

It is clear artificial intelligence can impact the integrity of the student and the workplace. With this comes the challenge of how we use it properly. It is silly to not think or try to stop students or workers from using it.

In my personal opinion, the best way to avoid academic dishonesty and cheating is to be diligent in avoiding putting yourself in situations that are tempting. That's a skill that will carry over to the workplace to continue to keep a person out of the hot seat.

Students' thoughts reveal that the use of Gen AI is inevitable, and we need to be prepared for tempting situations for dishonesty. Therefore, it is essential to find strategies using the tool with honesty and integrity in both academia and in the workplace.

Shared Responsibility to Address Implications of Gen AI

While some students placed the impedance of action on individuals (5%) stating that individuals needed to “make their own judgments” or determine their own ethical position, it was more common (25%) for students articulated a need for society to work together to address the impacts and determine how to appropriately integrate Gen AI. Specifically, student calls for collaborative approaches noted the need to examine ethical use of Gen AI in academic settings and develop policies which clearly articulated when Gen AI could be used and for what purposes.

“Students need to be aware of academic dishonesty and professors and institutions need to create environments that relieve the pressure of dishonesty.”

“I am not convinced that academic dishonesty is the same now as it was ten years ago. The world is ever changing. I believe policies and standards must change as well.”

While they were in the minority, students noted both the urgency to develop new standards to address Gen AI based cheating and the need to create balanced policies that did not entirely preclude the use of Gen AI.

Despite their call for collaboration on both policy and practice, most students did note that the ultimate responsibility for maintaining academic integrity lies with the student.

“Access to certain things on the internet can cause temptation to be dishonest or break workplace ethics.”

That is why it is important to have a strong understanding of workplace ethics and the potential consequences that result when not followed.”

“Ultimately, it falls to the individual to have the academic integrity and ethical foundation to truly learn the material and meet expectations or fail when presented with a problem that AI cannot solve.”

When specifically addressing how these violations could be avoided, most students articulated a need to rely on personal ethical foundations to choose not to engage in academic dishonesty, particularly given the ease of access provided through current iterations of Gen AI.

Discussions

Through the writing assignment, students were exposed to current research on the use of Gen AI in education and the workplace. While students were more likely to articulate threats than benefits, most students noted the coexistence of both. The results align with previous research that seeks strategies to maximize the benefits of Gen AI while minimizing the risks to academic integrity (AlAfnan et al., 2023; Fitria, 2023). Students also articulated an understanding of the different but interrelated potential uses and risks of Gen AI in areas beyond their academic careers in ways that mirrored the current discussions in society at large and those cited within the emerging literature (e.g., Fitria, 2023; Tlili et al., 2023). The parallels between the concerns of scholars (Davis et al., 2023) and those of students including academic dishonesty, loss of critical thinking skills, and generalized learning loss indicate a potential for constructive and cooperative efforts by students, faculty, and scholars to frame ethical standards for Gen AI.

Students expressed significant concerns about the use of Gen AI for cheating. Their concerns ranged and differed some between majors, but included poor future outcomes when learning is cut short, concerns about fairness and equality, and concerns about the erosion of both motivation and creativity when traditional understandings of academic integrity can be easily circumvented. These concerns parallel academic integrity concerns expressed by scholars in current literature (AlAfnan et al., 2023; Tlili et al., 2023; Fitria, 2023), indicating that some uses of Gen AI particularly those that are strongly associated with commonly recognized academic dishonesty may be integrated easily into existing frameworks of academic integrity.

The nursing student body consists of many non-traditional and returning students seeking a second career (Dos Santos, 2020). Nursing students' critical perspectives on their future workplaces may be influenced by their prior work experience in the medical field as healthcare professionals (e.g.,

certified nursing assistants, licensed practical nurses, paramedics) who are joining the four-year nursing program to become registered nurses. Nursing professionals uphold high standards of ethical judgement and decision-making skills (Yeom et al., 2017) and have particularly expressed concerns related to their field of study in the healthcare sector. The findings revealed that nursing students identified Generative AI as a potential threat to their future workplaces, as it could violate patient data privacy and provide inaccurate information on relevant research. While nursing students often have more defined career paths as registered nurses, students in the humanities and business fields may lack a concrete vision of their future careers upon graduation. The certainty and prior work experience in the healthcare sector may have given nursing students a critical perspective on ethical issues in their future profession.

Students seemed open to clearly articulated standards for Gen AI use but also help expectations for transparency of Gen AI by the institution and faculty. These findings indicate that some students want institutions and faculty to set clear ethical boundaries for student use of Gen AI in assignments and want to actively engage in defining these new standards. The strong articulation of this shared role indicates that actively engaging students in the development of these policies in the future could lead to better acceptance and adherence.

A small but relevant number of students expressed concerns about the equity of the use of Gen AI between faculty and students. These students articulated a perceived unfairness of institutional and instructional uses of Gen AI when they perceived student use would be prohibited. These responses indicated a desire for shared transparency in the use of Gen AI in academic settings where accurate disclosure of when and how Gen AI is used might allow for broader use by both instructors and students within the teaching and learning process.

Variance between different likely major groups indicates that student use and perceptions of Gen AI may differ by discipline. Additional research is needed to better understand how students in different disciplines view Gen AI including motivations or concerns regarding the threats and benefits of Gen AI in their academic and workplace environments. This indicates that different approaches to developing and articulating standards for student use of Gen AI may be needed that consider the unique uses, constraints, and opportunities for Gen AI across disciplines. For example, business and nursing students articulated different perceptions of the benefits of Gen AI for improving the educational and workplace experience. Conversely, approximately one-third of students across all majors expressed concerns about how the use of Gen AI might negatively impact human interaction indicating that students did not vary significantly across majors on all elements. Better understanding how students in different content areas view Gen AI could assist in determining where

differentiation in interventions to reduce unethical use of Gen AI would be the most useful and which methods were best suited to which student groups.

Limitations of the Study

The one-time nature of this data collection limits the scope of its applicability to address actual student behavior patterns. The nature of the document review using papers submitted to instructors at an institution with a clearly negative view of academic dishonesty likely limited the expression of fully divergent opinions related to ethical use of Gen AI in an academic setting any impact on student behaviors. Although attempts were made to provide balanced perspectives in the articles chosen, the provision of a list of possible articles for student use related to Gen AI may have influenced the opinions developed and expressed by students in their essays and thus the findings of this study. The intent was not to feed students with answers; however, was to give an opportunity to optionally select one of the articles provided if they were unable to successfully search a rigorous peer-reviewed journal article. The incoming non-traditional online students participating in the orientation program were in the process of learning “how to use technology for online learning,” and also “search library resources online.” As they were still learning and did not have much experience with the library system and reading research articles, we provided a list of six articles that the researchers believed were good quality peer-review journals and would benefit the students.

Future Directions and Implications for Research and Practice

The findings of our study provide a comprehensive overview of how non-traditional students perceive the emergence of Gen AI, their reflections on how to ethically use it in their educational process and prepare to use Gen AI in their future workplace. Students articulated the need for shared responsibility and action to avoid threats to academic integrity and develop new standards that consider both the threats and benefits of integrating Gen AI into the classroom and workplace. Our findings provide fundamental information that can lead to further exploration of how students can become an active member in the university community to voice their concerns and thoughts towards promoting ethical use of AI and safeguarding academic integrity issues. Building upon this discovery, future orientations within the research context or upcoming programs developing orientation initiatives may consider revising essay assignments to better align with the specific needs of students.

Our findings reveal that students across different fields of study exhibit varying degrees of critical awareness regarding the ethical implications of AI in their university courses, personal learning, and prospective workplaces. This variation

underscores the need for future research to delve deeper into each student body, aiming to identify their unique needs, beliefs, and the specific resources required to cultivate them into ethically conscious professionals. By doing so, we can better equip them to navigate the complex ethical landscape of their future careers.

With the growing number of non-traditional students in the United States, many of whom are pursuing second careers, it is crucial for universities to provide robust support tailored to their unique needs. As this demographic continues to expand, ensuring their success becomes not just important but essential for the future of higher education and the workforce. Students also looked for institutions and faculty to help them understand appropriate standards for use of Gen AI in academic settings and to help them avoid pitfalls that could lead to longer-term negative consequences. Given this insight, practitioners that include instructors, instructional designers, and leaders in higher education may consider how to better inform students of the appropriate use of Gen AI and developing policies that are transparent and supportive toward students.

A recommended future study would follow students beyond their initial orientation experience to examine long term connections between options expressed in the orientation course regarding the ethics of Gen AI in academic integrity and student propensity to engage in academic dishonesty while completing their degree.

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